

Math walkthrough for $F(m, n)$ when $n = 2^k$

1 FLOP Count for each step

See the table.

2 Recurrence from the code

For $n == 1$ the code does one *larfg* on a vector of length m , giving

$$F(m, 1) = 3m$$

For $n > 1$ and n a power of two, the code splits

$$n_1 = n/2$$

$$n_2 = n/2$$

and the recurrence becomes

$$F(m, n) = F(m, n/2) + F(m - n/2, n/2) + C(m, n)$$

where the local flop contribution from the helper calls is

$$C(m, n) = n_1 n_2 (6m - 2n_1 + 1)$$

For $n_1 = n_2 = n/2$ this is

$$C(m, n) = \frac{n^2}{4} (6m - n + 1)$$

3 Key simplification: $F(m, n)$ is linear in m

We claim that for $n = 2^k$, the recurrence

$$F(m, n) = F\left(m, \frac{n}{2}\right) + F\left(m - \frac{n}{2}, \frac{n}{2}\right) + \frac{n^2}{4} (6m - n + 1)$$

with base case

$$F(m, 1) = 3m$$

has the form

$$F(m, n) = 3n^2 m + b(n),$$

where $b(n)$ is independent of m .

Operation	Function	FLOPS(n_1, n_2, m_3)	FLOPS($n_1 = n_2 = n, m_3 = m - 2n$)
Factor left panel	GEQRT3($m \times n_1, n_1 \times n_1$)	$F(m, n_1)$	$F(m, n)$
Update trailing matrix	TRMM(L, $n_1 \times n_1, n_1 \times n_2$)	$n_1^2 n_2$	n^3
	GEMM($n_1 \times n_2, n_2 \times n_2, n_1 \times n_2$)	$2n_1 n_2 n_2$	$2n^3$
	GEMM($n_1 \times m_3, m_3 \times n_2, n_1 \times n_2$)	$2n_1 n_2 m_3$	$2n^2(m - 2n)$
	TRMM(L, $n_1 \times n_1, n_1 \times n_2$)	$n_1^2 n_2$	n^3
	GEMM($n_2 \times n_1, n_1 \times n_2, n_2 \times n_2$)	$2n_2 n_2 n_1$	$2n^3$
	GEMM($m_3 \times n_1, n_1 \times n_2, m_3 \times n_2$)	$2m_3 n_2 n_1$	$2(m - 2n)n^2$
	TRMM(L, $n_1 \times n_1, n_1 \times n_2$)	$n_1^2 n_2$	n^3
Factor right panel	DIFF($n_1 \times n_2, n_1 \times n_2$)	$n_1 n_2$	n^2
Construct T	GEQRT3($(m - n_1) \times n_2, n_2 \times n_2$)	$F(m - n_1, n_2)$	$F(m - n, n)$
Construct T	TRMM(R, $n_2 \times n_2, n_1 \times n_2$)	$n_1 n_2^2$	n^3
	GEMM($n_1 \times m_3, m_3 \times n_2, n_1 \times n_2$)	$2n_1 n_2 m_3$	$2n^2(m - 2n)$
	TRMM(L, $n_1 \times n_1, n_1 \times n_2$)	$n_1^2 n_2$	n^3
	TRMM(R, $n_2 \times n_2, n_1 \times n_2$)	$n_1 n_2^2$	n^3

Proof. We proceed by induction on n .

For the base case $n = 1$,

$$F(m, 1) = 3m,$$

which can be written as

$$F(m, 1) = 3(1)^2 m + b(1)$$

with $b(1) = 0$. Thus the claim holds for $n = 1$.

Now assume that for some power of two $n/2$,

$$F(m, n/2) = 3 \left(\frac{n}{2}\right)^2 m + b\left(\frac{n}{2}\right).$$

This is the induction hypothesis.

Substituting the induction hypothesis into the recurrence gives

$$\begin{aligned} F(m, n) &= \left[3 \left(\frac{n}{2}\right)^2 m + b\left(\frac{n}{2}\right) \right] \\ &\quad + \left[3 \left(\frac{n}{2}\right)^2 \left(m - \frac{n}{2}\right) + b\left(\frac{n}{2}\right) \right] \\ &\quad + \frac{n^2}{4} (6m - n + 1). \end{aligned}$$

Collecting the terms containing m ,

$$\begin{aligned} \text{coeff}_m(F(m, n)) &= 3 \left(\frac{n}{2}\right)^2 + 3 \left(\frac{n}{2}\right)^2 + \frac{n^2}{4} \cdot 6 \\ &= \frac{3n^2}{4} + \frac{3n^2}{4} + \frac{3n^2}{2} \\ &= 3n^2. \end{aligned}$$

Hence

$$F(m, n) = 3n^2 m + \left(2b\left(\frac{n}{2}\right) - \frac{3n^3}{8} - \frac{n^3}{4} + \frac{n^2}{4} \right).$$

The expression in parentheses depends only on n , so we define

$$b(n) = 2b\left(\frac{n}{2}\right) - \frac{5}{8}n^3 + \frac{1}{4}n^2.$$

Therefore

$$F(m, n) = 3n^2 m + b(n),$$

which completes the induction. □

That means for each fixed power-of-two n we can write

$$F(m, n) = 3n^2 m + b(n)$$

with $b(n)$ independent of m .

4 Recurrence for the offset $b(n)$

Plugging $F(m, n) = 3n^2m + b(n)$ back into the recurrence gives

$$b(n) = 2b(n/2) - (5/8)n^3 + (1/4)n^2$$

This is valid for $n = 2^k$ with $b(1) = 0$.

5 Solve $b(n)$ for $n = 2^k$

Define $n = 2^k$ and $g(k) = b(2^k)$. Then the recurrence becomes

$$g(k) = 2g(k-1) - (5/8) * 2^{3k} + (1/4) * 2^{2k}$$

Divide by 2^k and set $h(k) = g(k)/2^k$:

$$h(k) = h(k-1) - (5/8)4^k + (1/4)2^k$$

with $h(0) = 0$. Now sum over j where $1 \leq j \leq k$:

$$h(k) = -(5/8) \sum_{j=1}^k 4^j + (1/4) \sum_{j=1}^k 2^j$$

Use the geometric sums:

$$\sum_{j=1}^k 4^j = (4^{k+1} - 4)/3$$

$$\sum_{j=1}^k 2^j = 2^{k+1} - 2$$

So

$$h(k) = \left(-\frac{5}{8}\right) \left(\frac{4^{k+1} - 4}{3}\right) + \left(\frac{1}{4}\right) (2^{k+1} - 2)$$

Because $n = 2^k$, this is

$$h(k) = -\frac{5}{24}(4n^2 - 4) + \frac{1}{4}(2n - 2)$$

Simplify carefully:

$$h(k) = -\frac{5}{6}n^2 + \frac{1}{2}n + \frac{1}{3}$$

Then

$$b(n) = nh(k) = -\frac{5}{6}n^3 + \frac{1}{2}n^2 + \frac{1}{3}n$$

6 Final closed form

For $n = 2^k$, the flop count is

$$F(m, n) = 3mn^2 - \frac{5}{6}n^3 + \frac{1}{2}n^2 + \frac{1}{3}n$$

7 Base-case sanity check

For $n = 1$ this gives:

$$F(m, 1) = 3m - 5/6 + 1/2 + 1/3 = 3m$$

so the formula is consistent with the base case.

8 Numeric verification

I also verified this formula against the exact recursive count for powers of two:

1. $n = 2, 4, 8, 16, 32, 64$
2. and several m values ($m = n$, $m = n + 1$, $m = 2n$, etc.)

All matched exactly.

9 Summary

So the clean closed form for powers-of-two n is:

$$F(m, n) = 3mn^2 - \frac{5}{6}n^3 + \frac{1}{2}n^2 + \frac{1}{3}n$$